RIPARO.

The repair platform requirements baseline

Customer marketplace · Professional tools · Shop management · AI-assisted communication

Prepared for Anwar Hossain
18 October 2026 · Version 2.0

Explain the problem in your language. Choose an eligible professional. Approve the price. Follow the repair.

Review requirements first. The logical database and screen blueprints are proposals; final schema/migrations and the full visual screen set follow approval. This package does not certify existing software as production-ready.

Reading guide

The repair platformrequirements baseline	1
Reading guide	2
Riparo — requirements review package v2.0	3
Riparo — master product requirements	5
Technical team requirements and Codex delivery plan	16
Provisional logical database blueprint	24
Artistic design and screen blueprint	30
Riparo — investor and partner brief	35
Review decisions, sources and acceptance scenarios	38
Codex starter prompt	44

Use the editable Markdown and requirement register with Codex. This PDF is the human review copy. Proposed defaults, estimates and release targets remain labelled throughout.

Riparo — requirements review package v2.0

Prepared for Anwar Hossain · 18 October 2026 · Requirements baseline for review

What this package is

This is the proposed product contract for the Repairmen / Reparo / Riparo project. Riparo is the working product name throughout. The latest founder brief takes precedence over earlier documents. This package defines the full intended product, a bounded first release, functional behavior, technical delivery, a provisional logical database model, and the design brief.

The next decision is requirements approval. Production SQL, migrations and the final visual screen set follow that review. The logical model here is deliberately detailed enough to expose mistakes before schema implementation. No application has been built or deployed by preparing this package.

Read in this order

1. 01_MASTER_PRODUCT_REQUIREMENTS.md — product, journeys, requirements and business rules.

2. 02_TECHNICAL_TEAM_AND_CODEX_PLAN.md — architecture, delivery steps, API contracts and quality gates.

3. 03_LOGICAL_DATABASE_BLUEPRINT.md — entities, relationships, integrity and lifecycle rules; provisional.

4. 04_DESIGN_AND_SCREEN_BLUEPRINT.md — artistic UI direction and web/mobile screen inventory.

5. 05_INVESTOR_AND_PARTNER_BRIEF.md — opportunity, business model, illustrative economics and proof plan.

6. 06_DECISIONS_SOURCES_AND_ACCEPTANCE.md — review decisions, evidence, references and acceptance scenarios.

7. CODEX_START_HERE.md — copy this prompt into Codex with the package.

The PDF combines these documents for reading. The ZIP contains the editable Markdown originals and a machine-readable requirement register. Give Codex the ZIP contents rather than relying only on PDF extraction.

Main decisions captured

- Customer marketplace + professional workspace + repair-shop management + platform operations.

- Web/PWA launches first; customer and technician mobile apps reuse the same backend later. Design mobile-first from the start. The brief's phrase “first develop the mobile app” is interpreted as mobile-first UX preparation, because it explicitly states web launches before mobile. Native implementation order remains a review decision.

- Italy is the first operational pilot; Europe and the USA are expansion targets supported in the architecture. Proposed sequence is one Italian service area, then additional approved areas/countries, then a selected US city/state; simultaneous launch requires separately staffed country operations.

- Seven named trade categories, including retained Home Appliances, plus Other / I'm not sure.

- Speaking produces editable text. Voice notes are not the default communication object. Customer confirmation precedes dispatch.

- Technician response does not automatically book a repair. Customer selects one responding provider; a later quote requires separate approval.

- €9 per month per freelancer/provider organization or shop organization; not per employee. One-time €1–€4 acquisition fee for each new marketplace customer–provider organization relationship. Exact fee and tax presentation await review.

- Repeat work with that same organization carries no further customer-acquisition fee, even with another employee or branch. Repair payments and payment-processing costs are separate.

- PostgreSQL/PostGIS, Laravel modular backend, Next.js/TypeScript responsive web, Redis workers and object storage; React Native apps later.

- AI assists intake, transcription, translation and retrieval. Deterministic code controls eligibility, matching constraints, money, permissions, safety escalation and state transitions.

Review posture

Labels used throughout: **Founder requirement** = latest explicit request; **Retained** = prior decision supported by project records; **Proposed** = recommended default awaiting review; **Deferred** = full-product requirement scheduled after the first release. All targets and commercial scenarios are planning assumptions, not verified traction or guarantees.

Earlier status documents describe demos and source foundations, with real auth, payments, dispatch, background location and integrations still incomplete. Audit the actual repository before estimating reuse. Older files disagree on framework/database major versions; this package retains the stack but requires a tested, supported version matrix at kickoff.

Riparo — master product requirements

Version 2.0 · Review baseline · 18 October 2026

1. Product purpose and scope

Riparo helps a person explain a repair problem in their own language, find a suitable verified professional, agree the scope and price, follow the work and keep the records. Professionals receive understandable requests, manage work and retain customer relationships. Shops manage marketplace and directly acquired customers in the same operational workspace.

The product must work for older people, migrants, people with limited literacy and users with little technical knowledge. Speaking, taking a photo and confirming a short summary must be enough to begin. Forms and maps remain available without becoming prerequisites for describing a problem.

Product surfaces

Surface	Users	Purpose	Release
Public website	Visitors, prospective providers	Explain services, trust, languages, pricing and coverage; acquire requests	Web launch
Customer web/PWA	Customers	Intake, selection, quote approval, job timeline, payment, history	Web launch
Professional web/PWA	Freelancers and shop staff	Offers, availability, work execution, translated chat	Web launch
Shop management web	Owner, manager, reception, dispatcher, accounts	CRM, calendar, jobs, team, quotes, invoices and reporting	Core at launch; advanced modules later
Operations console	Support, verifier, finance, platform admin	Verification, dispatch exceptions, disputes, billing and configuration	Web launch
Customer mobile app	Customers	Native access, camera, push, voice and job history	After web pilot
Technician mobile app	Working technicians	Native offers, push, navigation, active-job location and work log	After web pilot

Full-product scope includes inventory, purchase/parts workflows, warranties, multi-branch operations and an inbound AI phone assistant. These remain specified but are phased; they must not block proving the core transaction.

2. Categories and service catalog

CAT-01. Use these stable category codes. Localized names can change without changing historical data.

Code	English	Suggested Italian	Examples
electrical	Electrician	Elettricista	Sockets, switches, lighting, circuits; licensed scope only
gas	Gas systems	Gas	Gas appliances and installations; safety screening first
heating_cooling	Heating & AC	Riscaldamento e climatizzazione	AC, heat pumps, boilers, heating, refrigeration
mobile_tablet	Smartphone & tablet	Smartphone e tablet	Phones, tablets, supported wearable devices
computer_printer	Computer & printer	Computer e stampanti	Desktop, laptop, notebook, printer, monitor, networking
plumbing	Plumbing & bathroom	Idraulico e bagno	Leaks, taps, drains, toilets, showers, sanitary fixtures
home_appliances	Home appliances	Elettrodomestici	Washing machine, dishwasher, fridge, freezer, oven, dryer
other_unsure	Other / I'm not sure	Altro / Non lo so	AI-assisted categorization, reviewed before dispatch

Home Appliances is retained because it is explicit in prior scope and covers frequent jobs omitted from the latest spoken category list. Refrigeration may be entered through Home Appliances, but qualification depends on the actual work: opening a refrigerant circuit differs from repairing an electronic control board. Heating may also involve gas; a category label alone never establishes legal eligibility.

CAT-02. Catalog hierarchy is category → subcategory → device/system type → optional brand → optional model. Brand is never a top-level trade. Examples: Computer → Laptop → Apple → MacBook; Appliances → Washing machine → Samsung → model.

CAT-03. A country/region/service-area matrix enables categories, modes, qualification requirements and schedules. Unsupported categories remain clearly marked unavailable, with a waitlist if enabled. “Other” routes to human triage when confidence is insufficient; it never bypasses trade checks.

CAT-04. Providers declare skills, work types, brands where relevant, exclusions, supported modes, travel area, hours, price basis and qualifications. A verified identity badge must not imply every trade is verified. Public badges explain precisely what was checked and when.

3. Roles, accounts and organizations

ACC-01. One human account can be a customer, an independent professional and a staff member, using an explicit workspace switch. Never mix permissions across workspaces. Customers may browse and draft a request before authentication; verify an account/contact before dispatch or payment.

ACC-02. A freelancer is a provider organization with one initial operator; a shop is a provider organization with staff and optional branches. The organization is the commercial counterparty, subscription owner and customer-acquisition fee owner. A worker is not a second billable shop solely because they receive an assignment.

ACC-03. Customer sign-in should begin with a simple OTP method; email/phone availability, anti-abuse cost and delivery determine the pilot choice. Support recovery, changing a verified contact, device/session revocation, account export and closure. Shop owners and privileged administrators require stronger authentication/MFA. Social sign-in is optional after the core flow works.

Permission contract

Role	Allowed scope	Restricted actions
Customer	Own requests, conversations, approvals, payments, documents	Cannot access another customer's records
Freelancer owner	Own organization, jobs, CRM and billing	Cannot access another provider's records
Shop owner	All organization functions; delegate roles	Platform verification/ledger manipulation
Shop manager	Assigned branches and operational management	Owner-only billing/security changes unless delegated
Receptionist	Customer records, bookings, allowed communications	Restricted margin data, refunds, staff roles
Dispatcher	Offer response, assignments, availability, routing	Price/finance access only if explicitly granted
Technician	Assigned jobs, relevant contact details, notes, draft quotes	Unassigned private jobs and organization-wide finances
Accounts	Invoices, approved payment/refund actions, reports	Trade verification and unrelated private media
Platform support	Cases and minimum necessary request information	No silent impersonation or unrestricted finance changes
Verification reviewer	Evidence and category eligibility	No automatic payment authority
Platform finance	Subscriptions, ledger, refunds, reconciliation	No changing repair facts or certifications
Super administrator	Explicit controlled platform operations	All sensitive actions audited; no shared admin logins

ACC-04. Staff invitations expire and grant only assigned branch/role access. Removing staff revokes sessions and job access; historical work authorship remains. Organization suspension stops new marketplace offers but must preserve lawful access to active job support and records under a defined policy.

4. Complete customer journey

CUS-01 — Start. Detect or ask display language; user can change it at any time. Ask city/postcode early to confirm serviceability without forcing exact GPS permission. Show one large “Tell us what needs fixing” action and all enabled categories. A category click preselects context but does not lock an incorrect diagnosis.

CUS-02 — Explain. Customer types or taps microphone, sees recording/transcription feedback, stops, edits the transcript and sends text. Photos and video are optional supporting evidence. Permission refusal must leave typing and file selection usable. Clearly show when recording is active. A failed recording offers retry or typed input without losing the rest of the draft.

CUS-03 — Voice is text-first. Temporary audio may be securely sent to a transcription service because browsers do not consistently offer suitable local recognition. It is not posted as a playable chat voice note by default. Transcript, original language, confidence and user corrections are the durable intake evidence.

Proposed default deletes temporary audio within 24 hours of successful transcription; deployment retention policy must approve this value. Retaining audio for a separate evidence or call feature needs explicit purpose, notice and retention. Optional read-aloud is a separate accessibility action.

CUS-04 — Organize. AI produces a short user-language summary and structured fields: reported symptoms, category, device, brand/model if supported by evidence, urgency, possible hazards, required skills and missing questions. It distinguishes customer statements from inferred hypotheses. It must not invent a model, diagnosis or certification need from a low-quality photo.

CUS-05 — Clarify. Ask one question at a time, normally no more than 3–5 decision-relevant questions before offering human assistance or a technician diagnosis. Ask only missing details that change safety, eligibility, mode, timing or price confidence. Model number is optional when unknown; “I don't know” must be accepted. If the selected category conflicts with evidence, explain the proposed correction and ask confirmation.

CUS-06 — Confirm ticket. Show a plain-language summary, thumbnails, service mode, timing, area and optional non-binding estimate. Customer edits and confirms. AI may create a draft ticket automatically; dispatchable submission requires customer confirmation, verified contact, safety clearance and serviceability. Record the confirmed revision and language. A meaningful change after submission invalidates affected provider responses and triggers reconfirmation/rematching.

CUS-07 — Choose mode. Offer only supported modes: home visit, pickup–repair–return, or customer drop-off. Explain any call-out, diagnosis, pickup, delivery and storage charges before acceptance. Changing mode recalculates availability and price disclosures.

CUS-08 — Timing. Offer now/today/scheduled, subject to actual coverage. Earlier urgent 15-minute and normal 30-minute matching windows are proposed search windows, not guaranteed technician arrival times. Scheduled work needs explicit start/end, timezone and provider capacity reservation. The customer can change time before booking; afterwards use rescheduling rules and mutual confirmation.

CUS-09 — Search and select. Explain that suitable providers are being contacted. Show responses as they arrive, normally no more than three at once. Each card shows organization/person, verified work scope, review count, languages, service mode, arrival/appointment estimate, price basis and response expiry. Customer may select one, decline, adjust preferences, wait for more or cancel. When a provider responds, request a permitted chime plus a visible banner and accessible announcement. The system cannot guarantee audible ringing when a browser/app is backgrounded, muted or restricted.

CUS-10 — Confirm booking. Choosing a provider atomically reserves one job and closes competing responses. Show any agreed diagnosis/call-out charge and cancellation policy before final confirmation. A second click/retry must not create another job. Customer can access translated chat and call options after booking; public offer cards do not expose the precise home address or phone number.

CUS-11 — Follow work. One job timeline combines provider confirmation, appointment, en route/ETA, arrival identity, diagnosis, quote, work, ready/return, completion, payment and support. Show last update time; stale location is labelled. The customer sees organization and assigned technician, and is notified when assignment changes.

CUS-12 — Approve quote. A diagnosis may lead to an itemized, versioned quote. Show labour, parts, diagnosis, travel, pickup/return, disposal, discount and applicable taxes separately. State which charges are already paid or credited. Customer explicitly approves a specific quote version and amount. If rejected, stop repair work and apply only previously agreed diagnosis/return charges. An increased scope requires a new version and approval; a chat message or AI response cannot substitute for the approval action.

CUS-13 — Complete and pay. Technician submits work summary and useful completion evidence. Customer confirms or reports an issue. A disputed completion cannot silently become an approved charge merely because a timer expired; any proposed timed process must be disclosed and reviewed. Online card or supported country-specific methods use a payment provider. Cash/external payment is recorded as such with a receipt where supported and a clear dispute policy. Payment failure does not erase job evidence.

CUS-14 — Aftercare. Keep approved quotes, invoice/receipt, warranty information, payment state and job history. Permit verified-job review, support/dispute and rebooking the same organization. The product must distinguish technician warranty promises, statutory rights and any platform guarantee. Do not advertise a guarantee that has not been funded and legally defined.

5. Professional journey and requirements

PRO-01 — Onboard. Choose freelancer or shop, create business profile, contact, service area, skills, hours, languages, modes and payout details. Submit identity/business/qualification evidence appropriate to location and work type. AI can extract fields; a reviewer or approved authoritative verification process decides eligibility. Rejected or expired evidence explains what needs correcting and provides appeal/support.

PRO-02 — Presence and availability. Provider can go online, offline, busy or scheduled-only. Presence heartbeats expire; an open browser alone is not proof of readiness. Job capacity, actual schedule, leave, branches and travel time control eligibility. Part-time providers are eligible during their declared hours without a blanket ranking penalty. Going offline stops new immediate offers while preserving active jobs.

PRO-03 — Receive an offer. Show translated symptoms, category, approximate area, mode, required skills, requested time, media permitted for the offer stage, estimate disclaimer, disclosed first-customer fee and response deadline. Hide precise address and sensitive customer details until booking. A provider can accept availability, suggest a time, decline with a reason, or report mismatch. Accepting availability is not an approved repair quote.

PRO-04 — Shop response. A permitted dispatcher may respond for a shop only if there is qualifying capacity. Final assignment requires a qualified staff member. A gas/heating job cannot be assigned to an unqualified employee merely because the shop has another qualified worker, unless the specific jurisdiction/supervision policy permits that work arrangement.

PRO-05 — Before arrival. Confirm appointment, route, disclose delay, communicate through translated text and optionally voice/video calling. Mark en route to enable limited active-job location sharing. Navigation can open a mapping app. Exact location sharing ends when the journey/job stage ends or is cancelled.

PRO-06 — At work. Arrival identity check, diagnosis, notes, photos, parts list, draft quote, customer approval, work log, checklists, labour time, extra-work request, completion and warranty. A customer-held arrival code is shown/said to the arriving technician and validated by the technician in the system; successful validation confirms the booked job. Rate-limit attempts; allow a support fallback when a customer cannot use the code. It is not an emergency authentication barrier.

PRO-07 — Money and history. Clearly separate repair earnings, customer balances, provider processing charges, platform subscriptions and one-time acquisition fees. Show gross revenue, refunds, recorded cost and contribution with definitions. “Profit” is not asserted if costs, taxes or staff labour are missing.

PRO-08 — Repeat customers. Rebooking remains with the organization. A customer may choose a preferred employee if available. No new acquisition fee for employee/branch reassignment within that organization. Changing legal ownership or splitting/merging organizations requires audited relationship migration and explicit fee treatment.

6. Matching and dispatch

MAT-01 — Hard eligibility first. Filter active country/area/category/mode, verified scope and valid qualifications, suspension/block rules, availability/capacity, travel feasibility, appointment compatibility and customer/provider safety exclusions. Mandatory constraints cannot be bought, waived by an LLM or overridden through ranking.

MAT-02 — Proposed organic scoring. Score normalized values on 0–1 using the following initial weights, versioned and configurable by area/category. This is a hypothesis to validate, not a trained optimal algorithm.

Factor	Weight	Definition
Relevant skill and device fit	25	Verified scope and demonstrated relevant work
Travel/arrival feasibility	20	Mode-specific ETA or route feasibility, not distance alone
Reliability	15	On-time arrival, completed commitments, upheld cancellations
Quality	15	Verified-job ratings with sample-size smoothing
Current responsiveness	10	Recent valid response behavior during declared availability
Shared language preference	5	User-chosen communication preference; translation remains available
Fair opportunity	10	Exposure opportunity among eligible, ready providers

Score = sum(weight × normalized factor). For each scoring version specify normalization, evidence window, missing-data defaults and ties. New providers start with neutral reliability/quality priors; they must not be permanently buried for having no reviews. Proposed prior: quality = (10 × area-category baseline + sum of verified ratings)/(10 + rating count), normalized to 0–1. Baseline defaults to neutral when local evidence is absent. Freeze scoring inputs per dispatch round for reproducibility.

MAT-03. Do not use nationality, ethnicity or culture as matching or price variables. Shared language is optional and user-controlled; do not infer nationality from language. Do not penalize a professional simply for being part-time. “Always online” earns no direct reward without honest availability and response quality.

MAT-04 — Commercial influence. The requested “who pays more” factor is handled as a separate future sponsored-placement module, clearly labelled and subject to eligibility, distance and quality floors. Baseline €9 subscribers use the organic matcher. No hidden pay-to-win weighting; no sponsored launch until pricing, disclosure and country review are approved.

MAT-05 — Dispatch rounds. Proposed urgent starting parameters: contact up to five suitable provider organizations, require a response within 90 seconds, retain availability responses for five minutes and display the best three. Expand search at configured intervals until the 15-minute search window ends. Normal jobs may search up to 30 minutes; scheduled jobs use explicit response deadlines rather than repeated ringing. These values are configuration defaults for review and may vary with geography. One shop receives one offer even if multiple employees qualify.

MAT-06 — Atomic booking. Provider response reserves capacity temporarily under an expiring hold. Customer booking rechecks eligibility, response expiry and capacity in a transaction. Use a database guard for one active booking per request and lock the relevant capacity reservation. If two customers select the same worker/time, only a valid available slot may commit. Other offers close only after the booking commits. Expiry, decline, cancellation and disconnection release holds idempotently.

MAT-07 — Fairness and recovery. Reserve a proposed 10% exploration share across eligible new/underexposed providers, still meeting qualification and ETA floors. Cap simultaneous offers, prevent repeated spam to a declining professional, log exposure and reasons, and monitor by area/category and declared availability. Lack of work is not enough to outweigh customer suitability. If no response, offer expansion of time/radius, a different mode or human assistance; never invent an available technician.

MAT-08 — Explainability. Store eligible candidate set, exclusion reasons, scoring inputs/version, round, ranking, response outcome and assignment result with restricted access. Public explanations use simple

relevant reasons without leaking private data. A human dispatcher can intervene with a recorded reason while respecting hard constraints.

7. Communication and translation

COM-01. Preserve original submitted text and source language. Generate translations per target language and source revision. Display translated text, a translation indicator and a “View original” action. Corrections invalidate old translations. Original text is never silently overwritten by a polished summary.

COM-02. Preserve numbers, currencies, addresses, brand/model identifiers, part numbers, dates and quote line items. A structured approval/price object is authoritative; translated prose is explanatory. Uncertain critical translation prompts reconfirmation or a human interpreter/support route.

COM-03. Launch languages proposed: Italian, English and Bangla, with Spanish prepared for the US pilot. Architecture supports BCP-47 language tags and RTL. Do not promise every language has passed QA; unsupported/low-confidence speech offers typing, another supported language or assistance. Customer and provider independently choose UI language.

COM-04. Text chat, media, delivery/read status, reporting and appointment events are core. Voice input in chat also becomes editable text before send. Optional calls are separately scoped: ordinary audio call first; live speech translation and video later. Call recording is disabled by default; country-appropriate notice/consent and retention are prerequisites for enabling it.

COM-05. Before booking, questions should use a mediated, limited offer Q&A with no exact address/contact disclosure. After booking, participants and authorized shop staff can communicate. Blocking/reporting stops abusive messages without destroying job records or the support path.

8. AI behavior, RAG and pricing estimates

AI-01. Use a provider-neutral AI gateway. Separate transcription, extraction, translation, retrieval, vision and user-facing phrasing. Each job records model/version, prompt version, input revision, tokens or processing duration, latency, validation result and cost estimate. Do not store unnecessary personal prompt content in analytics.

AI-02. A structured intake response must conform to a versioned JSON schema. It includes reported facts, hypotheses, evidence links, confidence, missing fields and next question. Reject invalid results, allow one bounded repair/retry, then fall back to simpler UI or human triage. The AI cannot execute arbitrary tools, SQL or administrator actions.

AI-03. Treat uploaded documents, OCR and RAG text as untrusted evidence. Instructions embedded in a photo/manual/message cannot change safety rules, billing, permissions or tool access. Retrieve only from approved, rights-cleared sources and tenant-scoped internal records. Never index one shop's CRM into another shop's search context.

AI-04. RAG stores source, permissions, locale, equipment scope, version, effective dates, confidence and retrieval citations. Start with approved troubleshooting summaries, manufacturer/service information permitted for use, qualification policies and local price data. If evidence is missing, say so. RAG is retrieval, not an internet scraping licence and not proof of diagnosis.

AI-05 — Price engine. LLM identifies plausible work and required questions; deterministic pricing code computes ranges from local historical jobs and curated price rules. Inputs include country/city, currency, equipment/work type, labour, diagnostic/call-out, travel, parts ranges, urgency, mode and tax policy. Return low/high, inclusions/exclusions, confidence, sample size, source freshness and expiry. A wide or absent range is better than invented precision.

AI-06. Example only: a supported local dataset might return €30–€50 for a defined minor job. This must not be presented as a default for all repairs. If parts/model/inspection are unknown, explain that the technician will diagnose and quote. US estimates need state/local rules and USD price data, not an EUR-to-USD conversion of Italian labour rates.

AI-07. No inference of guaranteed fault, technician qualification, payment completion or emergency safety. The actual quote comes from the provider; the customer approves. AI estimates do not authorize work, create binding price commitments or charge money.

AI-08. Local low-cost models handle suitable routine tasks after benchmarking; paid Gemini/OpenAI adapters handle approved escalation/vision when local results fail quality or latency gates. No assumption that self-hosted is free or that every request should call multiple paid models. See the technical plan for candidates, budget rules and a test set.

9. Service mode operations

OPS-01 — Home visit. Capture access instructions privately: floor, lift, doorbell, accessibility, parking where relevant. Notify assignment and arrival changes; provide identity/code checks, quote approval and support. A location permission refusal permits a typed address and map confirmation.

OPS-02 — Pickup/return. Create pickup and return appointments, named custody holders, item identifier, serial if available, accessories, condition photos, acknowledgement, transport responsibility and status at each transfer. A customer sees where the item is and who holds it. Failed pickup, inaccessible address, damaged/lost item, declined quote and return-only costs have explicit workflows and support escalation. A completed bench repair is not final delivery.

OPS-03 — Drop-off. Show shop address, opening hours and confirmed appointment/instructions. Generate a receipt or handover reference with item/accessories/condition. Track received → diagnosis → quote → repair → ready → collected. Collection uses a secure handover method and a delegate authorization path. Uncollected items use a reviewed notification/storage policy, never automatic disposal from an AI decision.

OPS-04 — Device privacy. Do not request device PINs/passwords in public descriptions or standard chat. If access is necessary, use a reviewed secure, time-limited process or customer-present unlock; minimize stored secrets. Photos can contain private screens/addresses: warn simply, restrict access and allow replacement/redaction where practical.

10. Shop management system

SHOP-01 — CRM. Customer list, verified/entered contact, communication language, consent/preferences, addresses, devices, history, notes, warranty and open balances. Distinguish marketplace-acquired customers from shop-owned/walk-in contacts. A local CRM contact does not automatically become a platform account or receive marketing messages.

SHOP-02 — Manual jobs. Reception can create a phone/walk-in job, link an existing contact, identify a device and use the same quote/work/payment timeline. Manual shop acquisition has no marketplace acquisition fee. If a customer account is later linked, require verified authority/consent and preserve provenance.

SHOP-03 — Calendar/dispatch. Day/week schedule, worker skills/hours, assignment, absence, travel buffers, multi-worker visits, appointment change notifications and collision prevention. The organization remains accountable for customer communication. Large jobs can have multiple staff but one provider counterparty and quote owner.

SHOP-04 — Quotes/documents. Draft, send, revision history, approval, conversion to job, deposits where supported, invoice/credit note/receipt, export and warranty. Store business/legal/tax identity snapshots with issued documents. Issued invoices cannot be silently edited; corrections use appropriate document

processes.

SHOP-05 — Inventory, deferred. Parts catalog, supplier, SKU/part number, quantity by branch/location, reserved parts, purchase/receipt, consumption against job, returns and stock adjustments. Prevent overselling reserved parts. Keep movement history rather than only editing a balance. Reorder alerts follow configured minimum stock. Integrations and supplier automation come after a reliable internal ledger.

SHOP-06 — Financial view. Collected/outstanding repair revenue, refunds, parts cost, recorded labour, platform fees and margin. Reports are organization/branch scoped and currency-specific. Export for the accountant. Tax filing, payroll and audited accounts are separate integrations, not implied by showing a dashboard.

SHOP-07 — Multi-branch and plans. Core model supports branches and staff; the first interface may focus on one branch. Proposed €9 Core includes essential marketplace, CRM, scheduling, quotes and document history with published fair-use limits. Advanced inventory, telephony and additional capacity can become optional plans only after founder approval. Do not invent compulsory per-seat charges.

SHOP-08 — AI phone receptionist, deferred. Identify business and AI clearly; collect caller language/problem/contact, check availability, draft or confirm an appointment within delegated rules, send confirmation and hand off to staff. It must not promise unsafe service, approve quotes, invent capacity or charge money. Recording, telephony costs, interruption, emergency handling, spam and failed-call recovery need their own acceptance suite. AI minutes are metered and cannot be unlimited inside €9 without demonstrated economics.

11. Revenue, billing and payments

BILL-01 — Subscription. Founder requirement: €9/month for a freelancer/provider organization or direct shop organization. Proposed per organization, not per staff member. Store plan, effective price, currency, tax treatment, seats/branches/features and fair-use limits as versioned configuration. EUR is the approved baseline; US subscription pricing is a separate review decision. The application must disclose whether €9 is tax-inclusive before selling it.

BILL-02 — One-time acquisition. Founder requirement: €1–€4 once for each first marketplace customer–provider organization relationship. Proposed exact launch fee €2; not yet approved. Proposed charging trigger is the first completed marketplace job, after completion eligibility is established. This avoids charging several technicians who merely responded. Fee disclosure occurs before response/booking; the backend stores the actual policy version and amount. The founder may choose booking-time billing after reviewing cancellation/refund consequences.

BILL-03 — Relationship rules. Relationship key is customer identity + provider organization. First job with shop A → one acquisition fee for A; repeat at A or another A employee/branch → zero further acquisition fee; first completed job with independent B → one fee for B. Direct/walk-in imports → no acquisition fee. Multiple devices, addresses, categories and employees do not create a new relationship. Account recovery and verified merges preserve history.

BILL-04 — Concurrency and refund. Two first jobs finishing together must create one relationship and at most one charge. Separate a durable relationship record from append-only charge attempts and adjustments. Refunds create credit entries. A refunded acquisition fee does not silently delete the relationship or cause it to be charged again; exceptional reversal/re-eligibility needs an audited policy. Deleted accounts must not become a simple fee-avoidance mechanism; preserve only the minimal legally approved billing linkage, with disclosed retention.

BILL-05 — Collection. Prefer aggregate monthly invoicing of acquisition fees with the subscription to avoid card-processing overhead on €1 transactions. Record each qualifying event individually; aggregate only collection. If invoice payment fails, retry according to disclosed rules, notify owner and use a grace period. Do not hold customer repair funds or active safety communication hostage to a failed subscription.

BILL-06 — Repair payment. The customer's repair bill is separate from Riparo subscription/acquisition revenue. Use a marketplace-capable payment provider and verified provider onboarding where Riparo routes payments. Define merchant/counterparty, refunds, disputes, payout timing and fee bearer before launch. No platform repair percentage is included in the founder's baseline model. Processor/FX/dispute fees still exist and must be allocated transparently. Do not build an unlicensed internal escrow wallet.

BILL-07 — Monetary integrity. All stored amounts have currency and exact minor-unit/decimal representation; never floating-point money. Historical quotes, taxes and invoices use snapshots. Payment webhooks are authenticated, deduplicated, order-independent and reconciled. A browser success page is not proof of payment. Refunds cannot exceed captured refundable balances.

12. Platform operations, safety and disputes

SAFE-01. A deterministic multilingual hazard screen runs before and during intake and when new evidence changes risk. Potential gas leak, fire, live shock risk and dangerous flooding interrupt ordinary booking. Show locally approved guidance and the appropriate official emergency route. Do not imply a repair technician substitutes for emergency responders, or that tapping acknowledgement proves the property is safe. Resume normal repair handling only through the approved incident policy.

SAFE-02. The console supports verifier queues, evidence expiry alerts, abuse reports, reassignment, no-show, cancellations, incident cases and disputes. Each action has actor, reason, timestamp, affected revision and audit event. Support access is purpose-limited and logged; exports and high-impact finance actions need extra control.

SAFE-03. Verified reviews come from actual completed jobs; allow a provider response and moderation/appeal. Prevent repeated self-review, review purchase and threats. Review removal follows a recorded policy rather than payment level. Handle customer and provider no-shows with documented evidence and previously disclosed fees.

SAFE-04. Dispute records contain issue, evidence, deadlines, communications, decision and financial adjustments. Customer sees a clear support path for damage, disputed price, missing item, unsafe conduct or poor repair. Refund, warranty return and rework are separate outcomes. Rework for the same relationship never triggers a new acquisition fee.

13. Global configuration and accessibility

LOC-01. Store country, region/state, city/service zone, postal code, timezone, language and currency explicitly. Country activation is an operational switch, not inferred from IP. Customer may change location; IP is only a suggestion. Addresses accept local formats. Store UTC instants plus scheduling timezone; display dates/times locally and handle daylight-saving transitions.

LOC-02. Support localized content and fallbacks, RTL layout, decimal/number formatting, kilometer/mile display and tax presentation. A request, quote, job and invoice retain the relevant country/currency snapshots. Do not sum mixed currencies into one earnings number without an explicit FX view and dated conversion rate.

LOC-03. Target WCAG 2.2 AA for web, with keyboard/screen-reader testing, visible focus, text alternatives, contrast and dynamic text. Use at least 48px practical touch targets, plain-language labels, text alongside icons, clear error recovery and no color-only status. An elderly person must complete intake without navigating a dense dashboard. Voice input, read-aloud and assisted service are options, not mandatory steps.

LOC-04. Browser restrictions must be visible in the plan: background location and ringing are limited on PWA. Native technician app is required for reliable supported background tracking after permissions/platform review. Web pilot promises foreground tracking and explicit status updates with fallback notifications, not continuous Uber-grade background operation on every phone.

14. Release boundaries and acceptance

Release A — managed Italian web pilot

One selected service area with approved providers and real operational support. All eight catalog entries are modeled; only staffed and approved work scopes are bookable. Core PWA journeys, transcription-to-text, translated text, hard-filter matching, customer selection, quote approval, three mode data models, home/drop-off operations, controlled pickup if staffed, payments, basic shop CRM/calendar/documents, billing and operations console. If pickup custody is not proven, show it unavailable rather than simulating it.

Release B — dependable web operation

Improve matching from evidence, automate reconciliation, introduce inventory/multi-branch features, deepen quote/tax-document integrations, extend language QA and add operational regions. Real scheduled work, exception handling and customer support must pass before expansion.

Release C — native mobile

Separate customer and technician apps, same API/business rules, native push/camera/voice, approved active-job background location and device-specific accessibility. No second independent billing implementation in the apps.

Release D — approved international expansion

Additional European markets and one selected US market after category licensing, tax, invoice, payment, insurance, consumer/privacy, emergency and support reviews. “Global architecture” is not evidence that every market can legally or operationally launch immediately.

Release E — AI phone and advanced services

Inbound AI receptionist, optional advanced plans, approved live speech translation, supplier integrations and measured sponsored placement. Enable only with adequate margins, consent and operational controls.

The requirement register and acceptance scenarios accompany this document. Core launch gates are: no cross-shop data access; no duplicate booking or acquisition fee; approved quote required; qualification enforced; translation preserves transactional facts; real payment reconciliation; no fake coverage; tested recovery from AI, queue, media and notification failure.

Technical team requirements and Codex delivery plan

Status: proposed implementation specification; review alongside the product requirements.

1. Architecture decision

Retain a Laravel modular monolith for transactional business rules, PostgreSQL with PostGIS for durable data and geography, Redis for queues/cache/short-lived coordination, and Next.js/TypeScript for public and authenticated responsive web. Add a small private Python AI worker/service for speech/model processing where useful. Customer and professional native apps use React Native after the web pilot.

This fits the founder's PHP/Laravel experience and avoids distributing the first booking and billing transactions across many services. A module boundary is a code ownership boundary; it does not require a separate server. Do not introduce RabbitMQ, Kubernetes, a second document database or a separate vector service without a demonstrated need.

PostgreSQL is the system of record. Redis loss must not erase confirmed bookings, billing or durable job events. Object storage holds media bytes; PostgreSQL holds authorization, ownership, revisions and metadata. pgvector is optional for approved RAG retrieval after verifying extension and embedding-model compatibility; ordinary relational/full-text retrieval is acceptable initially.

Proposed module map

Module	Owns	Key boundaries
Identity/organizations	Users, memberships, roles, sessions	Trusted actor and workspace resolution
Catalog/territories	Categories, work types, geography, policy	Country/category activation
Verification	Evidence, reviewer decisions, qualification scope	Eligibility read by dispatch
Intake/media/AI	Drafts, transcripts, summaries, temporary files	No booking or money authority
Dispatch	Candidate filters, rounds, responses, holds	No final quote authority
Jobs/scheduling	Booking, assignment, state, custody	One request → one active booking
Communication	Original messages, translations, participant access	Job/organization authorization
Quotes/documents	Versions, approval, invoice/receipt snapshots	Explicit customer price approval
Repair payments	Intent/capture/refund/payout/reconciliation	PSP-authoritative transitions
Platform billing	Subscription, relationships, acquisition ledger	Separate from repair funds
Shop operations	CRM, manual jobs, branches, inventory	Tenant isolation
Support/trust	Reports, disputes, incidents, warranties	Audited access and outcomes
Notifications/audit	Outbox, delivery attempts, append-only events	Retry and deduplication

Runtime topology

Browser/native → HTTPS edge → Next.js/web and Laravel API. Laravel communicates with the writer database, Redis and private object storage. Workers consume durable jobs and call approved AI/payment/message adapters. WebSocket/SSE events notify authorized participants; polling recovers state after disconnect. No browser receives provider secret keys or direct database access.

Use one PostgreSQL writer initially. Read replicas may later serve stale-tolerant reports; booking, permissions and money use consistent writer reads. UUIDs avoid ID collision; they do not provide safe multi-writer replication.

2. Repository and team ownership

Suggested structure:

```
apps/web/
apps/customer-mobile/      # after web gates
apps/technician-mobile/    # after web gates
services/api/app/Modules/
services/ai/
packages/api-contracts/
packages/design-tokens/
docs/product/
docs/architecture/decisions/
docs/qa/
infrastructure/
```

One product owner approves requirements and price policy. One technical lead owns data integrity and contracts. Backend owns authorization/state/money. Web owns accessible mobile-first flows. AI owns model evaluations, not financial policy. QA/operations jointly own end-to-end acceptance. A reviewer with relevant local expertise owns market-specific trade/safety/legal activation. Small teams may combine roles but must retain named accountability.

Do not assume a prior demo is the production repository. First inspect code, manifests, migrations, tests, live integration adapters and deployment evidence. Produce a reuse matrix: verified reusable, adaptation needed, demo-only, missing. Prior status documents explicitly leave real integrations and operational checks unfinished.

3. API and event contract

Version public endpoints under /api/v1; describe them in OpenAPI before client implementation. Use authenticated actor resolution and server-side workspace membership. organization_id sent by a client is a selector to authorize, never proof of tenancy. All UUID resource lookups check ownership/participants.

Resource/action	Contract essentials
POST /intakes	Anonymous/verified draft with abuse limits; draft access token
POST /media/upload-intents	Purpose, MIME/size limits, scoped expiring upload URL
POST /intakes/{id}/transcriptions	Async job, transcript revision, language, retry state

Resource/action	Contract essentials
PATCH /intakes/{id}	Expected revision; prevent silent overwrite
POST /requests/{id}/confirm	Confirmed source revision, mode, address/time, policy acceptance
POST /requests/{id}/dispatch	Allowed only after confirmation and safety/coverage checks
POST /offers/{id}/respond	Provider organization, qualified capacity, expiry, no duplicate response
POST /responses/{id}/book	Idempotency key; atomic slot + request booking guards
POST /jobs/{id}/transitions	Explicit command, expected version, permitted actor, reason
POST /jobs/{id}/quotes	Itemized draft; immutable submitted version
POST /quotes/{id}/approve	Customer, exact quote version/hash, amount/currency
POST /jobs/{id}/messages	Original text/language and attachment permissions
POST /jobs/{id}/locations	Assigned technician, en-route state, time, accuracy, sequence
POST /jobs/{id}/payment-intents	Server-computed payable amount; PSP reference
POST /webhooks/{provider}	Signature verification, durable event receipt, deduplication
GET /workspaces/{id}/reports	Permission/currency/date scope and pagination

Mutation responses return resource version, canonical state and timestamps. Use stable machine error codes plus localized explanations: forbidden, stale_revision, offer_expired, no_capacity, quote_not_approved, payment_pending, qualification_expired. Use 409 for state/version conflicts, validation errors for bad inputs, and 202 for accepted asynchronous processing. Sensitive errors must not reveal that another tenant's resource exists.

Events include RequestConfirmed, OfferResponded, BookingConfirmed, AssignmentChanged, QuoteSubmitted, QuoteApproved, JobCompleted, PaymentUpdated, CustodyTransferred, RelationshipQualified, AcquisitionFeeRecorded. Version event schemas. Commit state and outbox row together; workers publish after commit. Consumers store processed-event keys. Design for at-least-once delivery and idempotent handling, not magical exactly-once network delivery.

Realtime payloads carry minimum data, resource version and event ID. Clients reconnect using cursor/last version and fetch canonical state. Authorize subscriptions to job/organization channels. Do not broadcast customer address or location on public provider channels.

4. State machines

Keep independent state dimensions. One giant status field cannot represent job, quote, custody and payment truth safely.

Domain	Normal progression	Exceptions and guards
Intake/request	draft → clarifying → customer_confirmed → searching → booked → closed	safety_hold, no_match, cancelled, expired; changed material facts require reconfirmation
Offer/response	sent → viewed → available/declined → selected/closed	expired; available creates bounded capacity hold

Domain	Normal progression	Exceptions and guards
Job	booked → scheduled → en_route/awaiting_dropoff → arrived/received → diagnosing → awaiting_quote_approval → authorized → in_progress → ready/completion_pending → completed	cancelled, paused, failed, disputed; work requires approved scope
Quote	draft → submitted → approved/rejected/expired/superseded	Submitted versions immutable; approvals exact-version only
Custody	customer → courier/technician → shop → courier/customer	Every transfer acknowledged; loss/damage incident linked
Payment	pending → authorized → captured → partly_refunded/refunded	failed, cancelled, disputed; PSP events authoritative
Subscription	trial_if_enabled → active → past_due → restricted/cancelled	Grace/access policy explicit; records retained appropriately

Drop-off skips en_route; home visit does not require shop custody. Cash payment uses an external-settlement record rather than a fictitious card capture. A support dispute overlays a job; it does not erase its earlier completion/financial history. Transition permissions and preconditions require an explicit table in code and tests. Manual overrides are audited and cannot bypass hard qualifications or silently charge a customer.

5. AI implementation and cost controls

Candidate stack to benchmark

Task	Initial candidate	Why it belongs / limit
Speech-to-text	faster-whisper with multilingual Whisper small/medium; int8 on CPU where suitable	Practical self-hosted speech baseline; test accents/noise and actual server latency
Structured intake and routine translation	Qwen3-4B or Qwen3-8B through a private inference runtime	Candidate local multilingual model; validate structured outputs, negation and each launch language
RAG embeddings	Qwen3-Embedding-0.6B with PostgreSQL/pgvector if enabled	Compact retrieval candidate; pin dimensions and version, rebuild indexes when changed
Photo/model-label interpretation	Approved paid vision-capable Gemini/OpenAI endpoint initially	Use only needed images; validate brand/model and uncertainty; avoid CPU-heavy vision promises
Price range	PostgreSQL price rules + historical-data service	Exact arithmetic, evidence and localized ranges; no LLM invention
Text-to-speech, optional	Browser/platform voice initially	Availability/quality varies; always retain readable text

Qwen and faster-whisper are candidates rather than claims of best/current-newest models. Review exact model/weight/runtime licences before shipping. Self-hosting cost includes hardware, power, monitoring and engineering. Existing CPU/RAM capacity can support a pilot benchmark but does not establish acceptable concurrent production latency. Isolate AI CPU/memory from the database and web workers. Start with bounded concurrency and measure.

The official faster-whisper repository documents CPU int8 execution; its benchmark is not a performance promise for this server [W1]. The Qwen3-8B and embedding cards are starting references for features/licences [W2–W3]. Do not deploy every available model at once.

Routing policy

1. Check permission, input size and request budget; strip unnecessary personal information.

2. Run deterministic hazard/validation rules and retrieval where useful.

3. Use a benchmark-approved local model for routine extraction/translation.

4. Validate JSON and confidence signals; one bounded retry or clarification.

5. Escalate to an approved paid adapter only when permitted and within a budget.

6. Fail to editable text/manual support when no valid result is available. Booking/billing continue through deterministic forms where possible.

Set proposed per-intake caps: five clarification turns, two paid calls total, configurable input/output tokens and a €0.05 average variable-AI-cost pilot target per submitted request. This target excludes hardware, support, messaging and storage and is not a provider price. Long chat translation is separately measured and subject to fair-use limits. Daily organization/global budget caps, circuit breakers, per-IP/account throttles, duplicate request cache and maximum image/video duration prevent abuse.

Use a configurable paid-provider price registry rather than hardcoded “latest” prices. Cost formula: input_tokens/1,000,000 × input_rate + output_tokens/1,000,000 × output_rate + media/audio/tools costs; add FX when rates are USD and budgets EUR. Thinking tokens and media charges follow the selected endpoint's billing. Example with assumed rates $0.50 input/$2 output: 2,000 input + 500 output tokens = $0.002 for one call, excluding all other costs. This is arithmetic illustration, not a quoted tariff. Gemini's official rate page also distinguishes free/paid data-use conditions; select approved production terms for personal repair data [W4].

Evaluation before enabling a model

Build an initial consented/synthetic 200-case test set across Italian/English/Bangla/Spanish, all categories, code-switching, noise, missing models, hazards and misleading/prompt-injection inputs. Include negation (“not leaking”), decimal amounts, dates and part numbers. Proposed pilot gates: schema-valid extraction ≥99%; critical numeric/quote field preservation 100% on the release fixture set; category accuracy ≥95% on labelled ordinary cases; no missed hard-rule hazard fixtures; every low-confidence case falls back safely. These are acceptance targets on the test set, not claims of real-world perfection. Human review of language quality is required. Record latency/cost percentiles and error slices by language, category and device. Re-evaluate before model/prompt changes.

6. Security, privacy and media

Use TLS, secure cookie/session handling, CSRF protection for cookie mutations, short-lived scoped mobile tokens, rate limits, OTP quotas and privileged MFA. Policies enforce organization and participant scope on every endpoint, export, background job and WebSocket channel. Use database row-level security as defense in depth where correctly integrated; application policy tests remain mandatory. Pooling/worker tenant context must be cleared between jobs.

Uploads use allowlisted MIME/extension signatures, byte/duration limits, malware screening, quarantined processing, EXIF/GPS stripping when appropriate, safe thumbnails/transcoding and private object storage. Proposed defaults: images up to 10MB each and ten per intake; video up to 60 seconds and 50MB; voice transcription up to two minutes. Limits are configurable and the user sees them before uploading. Finalize uploads server-side; do not trust a client-provided object key. Signed download URLs are short-lived and

authorized per access.

Encrypt sensitive data/backups, keep secrets in a secret manager/environment, redact logs and restrict production access. Implement consent/notice versions, access/export/correction/deletion workflows, legal holds and data-category retention. Deletion propagates to derivatives, translations, search indexes and media, with accounting/legal exceptions documented. Customer identity and tenant contacts require a lawful-linking design; no phone-hash “anonymity” assumption. Choose processing regions and approved vendor contracts before real personal data is sent to cloud AI [W6].

AI output is escaped like any other user input; tool access is allowlisted and least privilege. No agent can issue a refund, edit a bank account or override a quote based solely on a conversation. PSP-hosted payment collection avoids storing raw card details.

7. Infrastructure and operational requirements

Deploy web/API, queue workers, scheduler, realtime service, AI worker, database, Redis and object storage explicitly. Uploading code alone is not deployment. Configure domains/TLS, secrets, mail/OTP, payment webhooks, maps/routing/geocoding, storage CORS, workers/supervisor/systemd, cron/scheduler, backups, monitoring and alert recipients. Produce .env.example without real credentials and a tested deployment/runbook.

Scheduled work includes offer/hold expiry, scheduled reminders, certification expiry, temporary-audio/media cleanup, retry/reconciliation, billing cycles, retention/deletion, notification retries and backup checks. Commands must be idempotent and avoid overlapping execution. Workers need timeout/retry/backoff/dead-letter rules; poison AI/media jobs must not block booking/payment queues.

Proposed initial targets to validate under a defined load: monthly core API availability 99.5%; ordinary non-AI API p95 below 500ms; user feedback within one second for asynchronous operations; local extraction p95 under ten seconds after upload, otherwise show progress/fallback. Pilot load profile: 100 concurrent authenticated users, ten new intake submissions/minute, and 50 simultaneous en-route sessions. Re-estimate from real demand. No fake benchmark claim.

Use point-in-time database recovery where supported, encrypted object versioning/backups and regular restore exercises. Proposed RPO 15 minutes and RTO four hours require actual infrastructure and a successful restore drill; do not promise them beforehand. Alert on outbox/queue lag, webhook failures, payment mismatch, offer expiry backlog, database saturation, AI latency/cost and storage errors. Keep trace IDs across a request without logging private raw content.

8. Development order and gates

Schedule is a planning range for a small experienced team, dependent on repository reuse, credentials, policy decisions and scope. A dependable web pilot may take roughly 12–18 weeks; it is not a delivery guarantee. Some design and operations work can overlap, but dependent integrity work must remain sequential.

Step	Work and deliverable	Exit gate
0 — 1 week	Review this baseline; choose pilot area/categories, counterparty and fee trigger; audit existing source	Decision log and reuse matrix accepted
1 — 1 week	Accessible mobile-first wireframes; test intake/quote with representative users	Owner approves representative flow before all screens
2 — 1–2 weeks	Logical model review → physical schema, migrations, auth/tenant foundation, CI	Schema/invariants accepted; tenant isolation tests pass

Step	Work and deliverable	Exit gate
3 — 1–2 weeks	Catalog, onboarding and qualification review, availability	Ineligible workers cannot receive/perform restricted work
4 — 1–2 weeks	Media, speech-to-text, confirmed intake, language preservation	Edited transcript survives; no unconfirmed dispatch
5 — 1–2 weeks	Eligibility/scoring, offer rounds, holds and booking	Concurrency/expiry tests prove no duplicate booking
6 — 1–2 weeks	Timeline, translated chat, appointments, foreground tracking	Reconnect and stale-location behavior work
7 — 1–2 weeks	Quote revisions/approval, custody, completion	No unauthorized work; handover evidence survives failures
8 — 1–2 weeks	PSP payments, platform billing, refunds/reconciliation	Sandbox webhook and concurrent first-fee tests pass
9 — 1–2 weeks	Core shop CRM/manual jobs/calendar/reports and operations tools	Cross-workspace and manual-source billing tests pass
10 — 1–2 weeks	Usability, load, accessibility, security, restore and operator drills	Launch checklist passes with real integrations
11 — measured pilot	Controlled live area, human support, instrumented requests	Evidence supports expansion and native investment

The summed sequence varies with reuse and overlaps; use gate-based planning rather than treating every estimate as a contractual date. Native apps follow a separate estimate after API/web acceptance. Advanced inventory and AI receptionist have their own milestones.

9. Codex working contract

Provide the Markdown package, decision register and actual repository. Ask Codex to read local AGENTS.md, inspect code and report current facts before editing. User authorization for this package does not authorize a later production deploy or emailing anyone. In the development session, give explicit scope and deployment authority separately.

For each feature, Codex must produce: requirement IDs covered, migration/API/UI changes, meaningful tests, changed paths, run commands, known gaps and a demonstration. Keep work in reviewable increments; use an isolated branch/worktree when the repository needs it. Do not erase existing migrations or user changes. A “done” feature must have real integration evidence or be clearly labelled mock/demo.

First implementation prompt

“Read this requirements package and the repository instructions. Treat requirements v2 as the proposed baseline and flag conflicts. First produce a source/reuse audit and requirement traceability map; do not rewrite the application. After requirements approval, implement only Step 2 foundation with PostgreSQL/PostGIS migrations, organization-scoped authorization, CI and meaningful isolation tests. Preserve existing working code. Report actual commands/results and unresolved environment dependencies. Never claim a mocked integration is live.”

Feature prompt template

“Implement [IDs] from [document], honoring approved decisions [IDs]. Inspect existing code and reuse it. Define schema/API/state effects first, then implement backend and UI, then test the stated failure/concurrency cases. Preserve tenant isolation, exact money and audit events. Use adapters for external services, include configuration documentation, and label unavailable credentials. Stop this increment when its exit gate passes; update traceability and the changelog.”

Use Codex for migration/API scaffolding, bounded feature implementation, fixture generation, review and regression fixes. Humans still approve ambiguous policy, user experience, real-language quality, business pricing and regulated launch. Do not ask an autonomous agent to infer legal eligibility from a free-text certificate or to “build everything perfectly” in one unreviewable change.

10. Mandatory test and release evidence

Test real database transactions for double acceptance, capacity races, unique relationship charging and webhook replay. Run against PostgreSQL/PostGIS rather than relying on SQLite for spatial/constraint behavior. Test two organizations and multiple staff roles for every private resource path, including AI retrieval, exports and sockets. Test revised quotes, out-of-order events, location expiry, DST scheduling, cash reconciliation and deleted contacts. Use mocked adapters for deterministic automated tests plus separate PSP/storage/OTP sandbox checks.

Every release needs CI results, dependency/security checks, restore evidence, accessibility/device checks, approved country configuration and a rollback plan. Deploy additive/backward-compatible changes first, backfill safely, then enforce constraints after validation. Rollback code without assuming an irreversible data migration can be undone. Production fixtures must never impersonate real bookings.

Provisional logical database blueprint

This document precedes physical schema approval. It describes what the database must represent and protect. It is not executable SQL or a verified migration set. Existing 104-table/other schemas must be compared against these rules before reuse; table count is not a quality target.

1. Universal conventions

- PostgreSQL is the durable system of record; PostGIS provides geography and spatial querying.

- UUID primary keys; UUIDv7 may be generated in application code with a documented compatible fallback. Pin actual engine/extension versions before choosing database-specific defaults.

- Use UTC timestamptz for instants, plus IANA timezone and local schedule intent where needed. Recurring hours are local rules, not fixed UTC offsets.

- Monetary fields have currency and integer minor units or exact numeric with explicit scale. Quantities/rates use exact numeric; totals use currency rounding rules. Never use float for money.

- Tenant-owned records carry organization_id; participant-shared marketplace records have explicit ownership/access joins. Composite foreign keys or equivalent enforced guards prevent cross-tenant joins.

- Record creator, timestamps and row version on mutable aggregates. Submitted quotes, issued documents and ledger events are immutable records with adjustment/revision processes.

- Soft deletion is a workflow option, not a privacy deletion strategy. Retention, erasure and legal holds are separate explicit processes.

- Use enums/checks for stable internal states and catalog tables for changeable business lists. JSONB holds raw model evidence or flexible metadata, not a substitute for required searchable financial/relationship fields.

2. Entities and principal fields

Fields below are minimum conceptual content. All entities have an ID and appropriate created/updated metadata unless expressly immutable. Physical types, naming and indexes are finalized after review.

Group	Entities and essential data	Relationships / constraints
Global config	countries, subdivisions, currencies, languages, service_zones, category_market_rules	Zone has country/timezone/geometry; enable each work scope/mode explicitly
Identity	users, verified_contacts, auth_identities, sessions, consent_records	Unique normalized verified contact according to recovery policy; consent version/purpose
Provider	organizations(type freelancer/shop), branches, memberships, roles, role_permissions, membership_scopes	Membership unique user+organization; branch belongs to same organization
Customer	customer_profiles, customer_addresses, customer_devices	Platform customer identity independent of shop-local CRM contact
Catalog	categories, subcategories, work_types, brands, models, model_aliases, work_qualification_rules	Stable codes; catalog links support cross-category equipment safely
Provider capabilities	provider_services, worker_skills, provider_languages, service_areas, business_hours, absences	Distinguish organization eligibility from assigned worker scope

Group	Entities and essential data	Relationships / constraints
Verification	verification_cases, evidence_files, certifications, certification_scopes, verification_decisions	Subject organization/person, jurisdiction, issuer, validity, checked_at and reviewer
Availability	presence_sessions, availability_slots, capacity_holds, capacity_reservations	Heartbeat expiry; booking window and worker/organization capacity
Intake	intakes, intake_revisions, transcription_runs, extracted_facts, clarification_turns	Preserve original text, corrected transcript, source language and confirmed revision
Media	media_assets, media_attachments, processing_jobs	Owner/purpose/storage key/hash/MIME/bytes/status/retention; no public storage ACL
Requests	service_requests, request_items, request_locations, request_time_windows, safety_assessments	Request source, category/work scope, mode, country/currency, confirmed revision
Estimates	price_rules, local_price_observations, estimate_runs, estimate_components	Price source/version/sample/confidence/tax scope; do not overwrite old estimates
Dispatch	dispatch_rounds, match_candidates, provider_offers, provider_responses	Offer per round+organization; scoring version, reasons and expiry
Jobs	bookings, jobs, job_assignments, job_events, appointment_changes	Single active booking per request; revisions and actor-authorized transitions
Custody	custody_items, handover_events, custody_acknowledgements, transport_legs	Item condition/accessories/from/to/location/time/evidence and current holder projection
Quote	quotes, quote_versions, quote_lines, quote_approvals	Approval references immutable version/hash and customer; all amounts/currency snapshot
Work	diagnoses, work_logs, job_parts, completion_reports, customer_confirmations	Assigned actor, permitted state and quote scope
Communication	conversations, conversation_participants, messages, message_revisions, message_translations	Original language/text retained; translation unique revision+target language+version
Location	location_samples, active_tracking_sessions	Worker+job, en-route state, sampled_at/received_at/accuracy/sequence; short retention
Notifications	notification_preferences, notifications, delivery_attempts, device_tokens	Event+recipient+channel dedup key; never confuse delivered with read
Repair funds	payment_orders, payment_attempts, captures, refunds, disputes, payout_references, webhook_receipts	Provider external IDs unique; exact amount/currency, append-only financial events
Platform billing	plans, plan_versions, subscriptions, billing_periods, provider_customer_relationships, acquisition_events, billing_ledger_entries, platform_invoices	Relationship key customer+provider organization; price/policy version at event time
Shop CRM	shop_contacts, shop_contact_links, contact_notes, contact_devices, contact_consents	Tenant-local identity; verified link to platform customer optional and controlled

Group	Entities and essential data	Relationships / constraints
Shop operations	manual_job_sources, branch_schedules, supplier_contacts, inventory_items, stock_locations, stock_movements, part_reservations, purchase_orders, purchase_lines	Manual jobs use shared job engine; stock movement ledger and tenant-safe suppliers
Documents	document_sequences, issued_documents, document_lines, credit_notes, warranty_terms, warranty_cases	Issuer/country/series/period uniqueness; immutable snapshots and corrections
Trust/support	reviews, review_responses, reports, support_cases, incident_events, moderation_actions	Verified job constraint; case-limited evidence access
AI/RAG	ai_runs, model_configs, prompt_versions, knowledge_sources, knowledge_documents, knowledge_chunks, embeddings, retrieval_runs, ai_feedback	Access scope, source rights/version, model/dimension, costs and input revision
Reliability/privacy	outbox_events, consumed_events, idempotency_records, audit_events, privacy_requests, retention_policies, legal_holds	Transactional outbox; request hash conflict guard; append-only audit with controlled retention

This inventory covers the full intended product; deferred entities do not all need migrations in the first release. Partition volatile location/AI/event history only when evidence justifies it. Keep the first physical design as small as the approved release permits.

3. Critical relationship examples

One user has many memberships; one organization has many branches and members. One organization owns each shop contact and stock item. A platform customer can have many unrelated shop-local contact records without granting cross-shop visibility.

One intake has many immutable revisions. One service request points to its confirmed revision, mode, location and schedule. It can have many dispatch rounds and provider offers, but at most one active booking. One job belongs to the booked provider organization and has many staff assignments/events. Reassignment changes the worker, not the commercial provider or acquisition relationship.

One job has quote headers and multiple versions; an approval points to one submitted version. Work logs reference the approved scope/version where relevant. A revised quote does not delete the earlier approval. A repair invoice and a platform invoice have different issuers and economic purposes even when linked to the same job.

One relationship connects a verified platform customer identity to one provider organization. It records first marketplace acquisition provenance and current fee disposition. Many jobs can reference it; only one qualifying acquisition charge is allowed under the current policy.

4. Non-negotiable integrity rules

Invariant	Proposed database/application enforcement
One active booking per request	Unique partial index on request for active booking states; transaction and explicit lifecycle
One response per provider/offer	Unique provider offer/response key and command idempotency
No worker double booking	Explicit time ranges; exclusion constraint for single-capacity workers, transaction/slot locking for multi-capacity organizations
One customer acquisition per provider organization	Unique canonical customer+organization relationship; unique charge purpose+relationship key; transactionally record ledger/outbox
No quote approval drift	Approval FK to immutable version; amount/currency/hash snapshot; command checks expected revision
No job starts without approval	Transition service verifies valid quote/scope approval or explicitly approved pre-priced task; audit and tests
No cross-organization children	Composite IDs/FKs or validated scoped associations for branch/member/contact/job/stock relationships
No double webhook effect	Unique external provider+event ID; durable receipt before processing; monotonic/allowed financial transitions
No mutable issued financial evidence	Append-only issued rows, credit/adjustment entries; restricted write roles
No mixed-currency arithmetic	Currency compatibility checks at quote/job/payment/document boundaries
No confidential media leakage	Ownership/participant purpose checks before signing object access; quarantine until processing complete
No duplicate mutation on retry	Idempotency record unique actor/workspace+route+key, stored request hash and response

Do not rely solely on Redis locks. Locks can expire during processing; unique constraints and transactional state guards remain authoritative. Idempotency keys reused with different request bodies must reject instead of returning unrelated prior success.

5. Booking transaction

1. Resolve authenticated customer and load submitted request, selected response and relevant capacity row with locking.

2. Recheck ownership, confirmed revision, response expiry, qualification validity, mode and timing.

3. Convert an available hold into a durable reservation and insert active booking/job; database constraint rejects duplicate active booking.

4. Record selected organization/worker, initial price disclosures and policy version. Do not charge the later repair quote here.

5. Append job event and outbox notification in the same transaction; commit.

6. Close competing offers and release holds through idempotent events. On race failure return a clear refresh/reselect state.

If an offer promised only organization capacity, assignment must still resolve qualifying worker capacity before the appointment can be committed or before the disclosed assignment deadline. Do not allow indefinite overbooking hidden behind “shop accepted.”

6. Acquisition fee transaction

1. Accept one authorized JobCompleted eligibility event; source must be marketplace and policy must qualify the event.

2. Resolve canonical customer identity and booked provider organization; create/upsert the unique relationship.

3. Lock relationship; if already charged/exempt under policy, record no-charge reason and stop.

4. Insert acquisition event and ledger debit with immutable fee/tax/currency/policy snapshots and a unique relationship purpose key.

5. Mark fee disposition, append outbox record and commit. Monthly collection can aggregate these debits separately.

6. Credits/refunds append ledger adjustments; retain prior event and relationship. Replay has no second financial effect.

Contact merge must redirect both jobs and relationships through a reviewed canonicalization service. If merging accounts reveals duplicate historical fees, create an auditable adjustment rather than deleting ledger rows. Provider mergers/splits and customer account erasure require separate policies before implementation.

7. Geography and search design

Store service location and worker position as appropriate PostGIS geography/geometry with SRID consistency. Use spatial indexes and meter-based ST_DWithin candidate filtering where suitable, then route/ETA estimation for the shortlisted candidates. A straight-line distance is not a promised road arrival time. Model service areas as polygons and/or radius rules with explicit mode support. Drop-off ranks by shop accessibility/distance and opening times; pickup considers both legs.

Sensitive home coordinates are private. Public nearby-provider maps show approximate locations or areas; never leak precise worker/customer positions to anonymous users. Live samples include accuracy, time and current job scope. Discard stale/out-of-order points for display while retaining necessary audit under the approved policy.

8. Index, retention and migration plan

Prioritize indexes on tenant+status/time, request/booking state, offer expiry, worker/time overlap, relationship customer+organization, provider external event ID, quote version and conversation+sequence. Add GiST for spatial/time exclusions where applicable. Inspect query plans with representative data; do not index every column speculatively.

Proposed retention classes: temporary transcription audio ≤24 hours after successful transcription; short-lived unconfirmed drafts with a configured expiry; location samples short-lived after the job; financial documents per reviewed jurisdiction policy; support evidence per case/legal hold. Exact periods besides the audio proposal remain decisions. Privacy deletion must include object derivatives, AI logs, search chunks and caches, with documented billing/legal retention exceptions.

After requirement approval: audit old schema → map old tables to domains → identify conflicts → write migration ADRs → implement additive migrations and seed catalogs → test fresh install and supported

upgrade → generate ERDs and data dictionary → test backup/restore. Never silently run old SQL plus new migrations with overlapping incompatible definitions. Physical schema deliverable must include actual PK/FK/check/index definitions, ownership, retention and example queries, with migration tests on the chosen engine.

Artistic design and screen blueprint

This is the design contract for the next design stage. It is not a claim that every final screen has been drawn. Approve requirements and three representative flow screens before producing the full customer, professional and desktop screen sets.

1. Direction from the supplied work

The reviewed Repair App Wireframes(1).pdf, page 1, uses a hand-drawn wordmark, outlined conversation cards, restrained accents, simple phone frames, a ticket inside the thread and a compact map after matching. Its second alternative is map-first. Preserve the illustrated, human character and functional clarity. Adapt the old voice-message depiction to the new requirement: microphone input becomes editable text before sending. Do not copy old fee or emergency behavior without the updated product rules.

The earlier design handoff specifies warm paper #FAF7F2, ink #201D1A, trust green #2C7A62 and terracotta #C4622D, Bricolage Grotesque headings, Instrument Sans body and Bangla fallback. These are retained starting tokens. Verify contrast in actual component states; not every colored text/background combination is suitable for small text.

Use artistic line illustrations of tools, appliances and friendly service moments, with restrained paper-like shapes. Avoid photographic category tiles, glossy device renders, rainbow category grids and decorative gradients. Keep letterforms readable; use the hand-drawn treatment for wordmark/illustrations rather than critical long text. Artwork never substitutes for a button label or error explanation.

2. Experience structure

Customer: category/voice-first entry followed by a guided conversation. A clear ticket card, provider-response cards, quote card and job timeline live within or beside that conversation. The map appears when it helps explain travel; it does not dominate the first experience or expose precise locations before booking.

Professional: action-first home: availability switch, next appointment and incoming requests. Use larger cards with “Available”, “Suggest time” and “Decline”. Separate work execution from earnings/billing to avoid accidental financial actions.

Shop desktop: calm workspace with Customers, Jobs, Calendar, Team, Quotes/Documents, Payments, Inventory when enabled, Reports and Settings. A job detail page is the common spine for customer, device, timeline, staff, quote, custody and communication. Avoid a dense chart dashboard before basic tasks are easy.

Operations: queues and exception details, with explicit status, reason and audit trail. Show exactly why a qualification, match, payment or upload is blocked.

3. Layout and interaction tokens

Design at 390px mobile and verify 320px width, 200% text zoom and large system text. Desktop layouts cover common 1280–1440px workspaces. Use an 8px spacing rhythm, generous whitespace, 16–24px card radius, thin ink/neutral outlines and little or no shadow. Practical touch targets ≥48px, body text around 16–18px, primary action around 18px where fitting. Long translated labels may wrap; never truncate the meaning of Approve/Pay/Cancel.

Customer bottom navigation proposed: Home, Repairs, Messages, Profile; one prominent “New repair” action is always reachable. Earlier five-item navigation can be tested, but avoid two competing new-repair controls

on a small screen. Professional navigation: Today, Requests, Jobs, Messages, More. Web desktop uses persistent side navigation and clear workspace switch.

Critical action hierarchy: one filled primary action, a clear secondary action, and a separate destructive action requiring context. Buttons say what happens: “Confirm ticket”, “Choose this professional”, “Approve €84.00 quote”, “Pay €84.00”. A repair estimate must not appear as the price on a pay button.

4. Representative screen compositions

A. Customer home

Top: illustrated Riparo wordmark, editable service area, language control. Main: “What needs fixing?” with a simple line illustration and large “Tell us” microphone/text action. Then the eight named categories with readable labels and consistent line icons; “I'm not sure” remains obvious. Bottom: current job or last professional when relevant. No login wall before explanation.

B. Intake and ticket confirmation

One short AI prompt; original text bubble; editable voice transcript preview; camera/video controls. A confirmation card shows equipment/problem, mode, time/area and “Estimate unavailable” or an evidenced range. Always show “Edit” and “Confirm ticket”. Errors preserve entered text. The review screen labels uncertain model detection instead of implying certainty.

C. Professional responses

Visible “A professional is available” banner, permitted chime and up to three comparable cards. Card: organization, assigned/expected professional, relevant verification, rating/count, languages, time and disclosed call-out/diagnosis basis. Primary “Choose” action; expiry is understandable. Show no-match recovery when needed. Sponsored placement, if ever enabled, has its own label.

D. Job, quote and payment

Timeline shows next action; active map is compact and last-updated. Translated chat exposes original text. Quote includes itemized total and separate approve/reject actions. Work-started state follows approval. Completion and payment distinguish provider confirmation, customer acceptance and PSP pending/success. Documents and support remain visible after completion.

5. Customer screen inventory

IDs	Screens	Required variants
C01–C04	Welcome/language; service area; home/categories; category detail	Unsupported area/category, returning customer
C05–C09	Intake text; voice transcript review; photo/video upload; clarification; safety interruption	Permission denied, upload/transcription failed, uncertain language/model
C10–C14	Ticket review/edit; OTP/account; mode selection; timing; address/access	Unknown model, no GPS, unavailable mode, timezone conflict
C15–C19	Searching; responses; provider profile; booking confirmation; no-match recovery	Expired response, lost capacity, cancellation
C20–C24	Job timeline; map/ETA; translated chat; call handoff; arrival check	Stale location, translation unavailable, changed technician
C25–C29	Diagnosis; quote detail; quote approval; revised quote; waiting/work state	Reject, expire, extra work, previously paid diagnosis credit
C30–C34	Pickup/drop-off handover; custody detail; ready/return; completion review; issue report	Missing accessory, failed delivery, damaged/lost item
C35–C39	Payment choice; PSP handoff; payment result; documents/warranty; review	Failed/pending/refunded payment, disputed completion
C40–C44	Repair history; rebook; profile/preferences; privacy/account; support case	Closed provider, account recovery, export/deletion pending

These are reusable screen/state families, not 44 compulsory separate pages in code. Shared flows may use drawers/cards where usability permits. Every screen needs loading, empty, validation, offline/retry and permission-aware behavior where relevant.

6. Professional and shop screen inventory

IDs	Screens / families	Notes
P01–P05	Type selection, business/identity, skills/modes, languages/area/hours, evidence upload/review	Document expiry/reject/resubmit
P06–P10	Today/availability, request list, request detail, response/time proposal, booking result	Expired offer and lost-slot states
P11–P15	Calendar, assignment, route/en-route, arrival validation, diagnosis/work checklist	Role/skill and location permissions
P16–P20	Quote builder, submitted quote, approved work log, parts/photos, completion	Revised quote blocks extra work
P21–P25	Custody handover, ready/return, translated chat/call, customer history, warranty/rework	Item/accessory evidence
P26–P30	Earnings/balance, platform subscription, acquisition-fee ledger, notifications, profile/support	Repair funds separated from platform charges
S01–S05	Shop overview, CRM list/detail, manual booking, device history	No marketplace fee for own acquisition
S06–S10	Dispatch board, team/roles, branch settings, calendar capacity, job detail	Multi-staff with one organization
S11–S15	Quotes/documents, payments/refunds, revenue/cost reports, export, settings/billing	Correct permissions and currencies
S16–S20	Inventory, suppliers/purchases, stock movements, warranty queue, AI receptionist settings	Later phase unless specifically enabled

7. Public web and operations inventory

Public pages W01–W12: home, service directory, category detail, approved area page, how it works, trust/safety, language support, professional recruitment, provider pricing, FAQ, contact/support and legal/privacy/terms. Additional account/request pages reuse the customer PWA. Public provider profile may be enabled only with consent and a defined publication policy. Index only real service areas and useful localized content; private tickets/chat/media stay out of search indexing.

Operations O01–O12: overview/alerts, verification queue/detail, provider/organization detail, request/match trace, job/incident detail, support/dispute, subscription/acquisition ledger, payment reconciliation, catalog/territories, policy/model configuration, audit/privacy requests and reporting. Permissions govern every view and action.

8. Prototype validation and handoff

First build Home → Intake → Ticket → Responses → Quote → Completion for one ordinary job; include transcript failure, no match and quote change. Test with 5–8 representative users including older, migrant and less technically confident people. Proposed task goal: most participants can submit a correct request and explain the approved price without assistance. Record actual findings, not a fabricated pass rate.

After review, deliver high-fidelity mobile customer/pro screens, responsive desktop shop/operations screens, reusable components/tokens, localized long-text examples, redlines, interaction notes and keyboard/screen-reader behavior. Native app screens follow the same approved journey, with documented platform push/location/camera differences. Final visual design is a separate deliverable after this requirements review, as requested.

Riparo — investor and partner brief

Review-stage product case · Not a fundraising valuation or verified financial forecast

1. Customer problem and proposition

When something breaks, many customers cannot describe the technical fault, judge the right trade or communicate comfortably in the local language. Repair businesses receive fragmented calls and messages, spend time clarifying requests and often manage quotes, parts and customer history in disconnected tools.

Riparo proposes one simple journey: explain the problem in your language, confirm an understandable ticket, choose an eligible professional, approve the price and follow the job. Shops keep the relationship and manage repeat work in their own workspace. This is a product hypothesis based on the founder's brief; customer demand and willingness to pay still need live pilot evidence.

2. Three priority value hypotheses

Workflow	Change	Expected benefit	Measurement / evidence posture
Describe and triage	Voice-to-text, photos, structured clarification	Less abandonment and fewer unsuitable dispatches	Completion rate, clarification time, mismatch rate; assumed until measured
Coordinate and authorize	Translated conversation, timeline, exact quote approval	Fewer misunderstandings and price disputes	Response/arrival time, quote approval rate, upheld disputes; assumed
Operate and retain	CRM, schedule, documents and no repeat acquisition charge	More organized repeat work and provider retention	Repeat usage, monthly churn, support minutes; assumed

Differentiation must be demonstrated in this combined workflow. Do not claim Riparo is the first, only or best AI repair marketplace, or that a model by itself guarantees superior matching.

3. Product and launch strategy

The full product has customer access, professional access, shop management and operations. A responsive web/PWA pilot establishes the transaction and supply quality before separate native apps. Proposed first operational market: one staffed Italian service area with approximately 15–25 approved provider organizations as a recruitment target. The final city/zone is not newly confirmed; earlier work recommended a Rome-area pilot. Expand to other approved European areas and a selected US city/state after operational and jurisdiction readiness.

The architecture supports multiple languages, currencies, regions and timezones. This is a global expansion design, not proof of immediate worldwide coverage. Early gas/electrical hazardous incidents route to approved emergency handling rather than an ordinary repair booking.

4. Founder-defined revenue model

Known from latest brief: €9 monthly per freelancer/provider organization or shop organization; €1–€4 once per new marketplace customer–organization relationship; repeat jobs with that organization have no new acquisition fee. Proposed exact fee €2 and first-completed-job trigger await review. The fee is paid by the provider, not by every technician receiving an offer. Repair bill value is separate from platform revenue.

Advanced inventory, higher capacity, telephony/AI minutes or clearly labelled sponsorship may create optional later revenue. These are unapproved extensions and excluded from the baseline below. There is no hidden repair commission in this model. US plan prices, tax inclusivity and processor-fee allocation remain open.

5. Illustrative monthly revenue scenarios

Formula: paying provider organizations × €9 + qualifying first-customer relationships × effective acquisition fee. Count a relationship once, not every ticket, employee or repeat repair. Values below are assumptions for sensitivity, not demand forecasts or existing traction. Use revenue excluding indirect taxes; if the advertised €9 includes tax, recognized net revenue is lower.

Scenario	Providers	New pairs / month	Fee	Plan revenue	New-pair fees	Total before costs
Small pilot	100	200	€1	€900	€200	€1,100
Working local network	500	1,500	€2	€4,500	€3,000	€7,500
Broader rollout	2,000	8,000	€4	€18,000	€32,000	€50,000

At the working-network assumption, acquisition revenue would be €1,500/€3,000/€6,000 at a €1/€2/€4 fee, producing €6,000/€7,500/€10,500 total with 500 subscriptions. The outer scenarios deliberately change both scale and fee; they are not comparable growth forecasts.

Repair GMV is the value of customer repairs and is not Riparo revenue. Customer funds owed to providers are not available operating profit. Refunds, discounts, failed collections and churn reduce realized revenue. Annualizing an illustrative month does not establish ARR for variable acquisition income.

6. Cost and sustainability

Main costs: engineering, support/dispatch, provider verification, acquisition/marketing, servers, AI, translation, media, notifications/OTP, maps, payment/FX/disputes, insurance/legal/accounting and charge/refund losses. No verified cost baseline or acquisition/churn evidence was supplied, so profit and break-even provider counts cannot be credibly promised.

At a pure planning assumption of €0.05 variable AI cost per submitted request, 10,000 requests consume €500 in variable AI; this excludes messaging, infrastructure and human handling. A six-minute support interaction at an assumed €15/hour costs €1.50, which can consume much of a €1–€2 acquisition fee. This is why support workload and fair use matter as much as token savings.

For a provider with an assumed three new charged relationships/month at €2 each: gross platform revenue = €9 + 3×€2 = €15/month before taxes, losses and costs. Contribution per provider = net recognized subscription + net acquisition revenue − allocated variable AI/messaging/verification/support/payment costs. Break-even providers = fixed monthly costs / positive average contribution per provider. Missing inputs: actual fixed costs, net plan price, variable workload, collection losses, CAC and churn.

Low price helps recruitment but leaves little room for unlimited live AI calls, media or manual support. Preserve the €9 baseline while publishing reasonable capacity/fair-use limits and offering optional metered costly features only after validation. Self-hosting needs measured total cost and latency, not a claim of free AI.

7. Pilot proof and investment decisions

Proposed evidence collection: first 100 qualified service requests, actual matched/accepted/completed jobs, two monthly subscription renewal cycles, and interviews with retained and departed providers. These are targets, not achievements. Track channel, request category/language, serviceability, response/selection, ETA adherence, price accuracy, refund/dispute, support minutes, total variable cost and repeat relationships.

Define qualified request consistently: customer-confirmed, serviceable, non-duplicate, verified contact, within enabled scope, and not an emergency referral. Define paying provider as collected subscription rather than merely registered. Retention uses cohorts; reviews and completion rates need denominators and sample counts.

Proposed expansion gates: reliable eligible supply during advertised hours; at least 70% of qualified pilot requests booked; at least 80% of booked jobs completed without unresolved operational failure; no unresolved critical safety/data/payment defect; positive measured variable contribution or an approved route to it. Small pilot samples cannot prove rare safety events are solved. These thresholds are initial management targets requiring review, not market benchmarks.

Funding should follow evidence gates: validate and recruit → implement trustworthy web transactions → run controlled live pilot → improve retention/contribution → invest in native apps and new markets. Budget and funding amount remain missing and should be built from the team's actual costs after repository audit. No valuation or investment return is asserted.

8. Partner roles and open evidence

Local repair partners supply qualified capacity, transparent prices, custody discipline and aftercare. Community/language partners can help usability and recruitment with consent-based outreach. Payment, insurance and document providers can close jurisdiction-specific operational gaps. The founder/team supplies the platform, onboarding, support and honest reporting.

Known: product brief, proposed pricing and earlier demos/source foundations. Inferred: the integrated workflow may reduce friction. Assumed: willingness to pay, conversion, retention and cost targets. Missing: audited deployment readiness, signed provider supply, market-specific activation approvals and real cohort economics. A credible partner presentation should show the product as proposed/demo until those facts are established.

Review decisions, sources and acceptance scenarios

1. Review decisions

The package is usable for review without answering these immediately. Resolve them before the corresponding implementation or market activation. Proposed defaults are recommendations, not founder approvals.

ID	Decision	Proposed default / current posture	Needed before
D01	Product name	Riparo working name; check ownership/domain/trademark separately	Public branding
D02	Launch order	Mobile-first design; web/PWA first; native apps after core web pilot	Project scheduling
D03	First geography	One staffed Italian area; prior work suggested Rome, not newly confirmed	Provider recruitment/coverage
D04	Europe/USA rollout	Architect for both; activate each approved market in sequence	Market launch
D05	Categories	Seven named trades including Home Appliances + Other/unsure	Catalog and UI
D06	Subscription unit	€9 per provider organization/month, not per worker	Billing schema/terms
D07	Fee amount	Configurable €1–€4; proposed €2 first-customer fee	Billing policy
D08	Fee trigger	First completed qualifying marketplace job	Billing policy
D09	Tax/country pricing	Decide €9 inclusive/exclusive tax; approve USD prices separately	Selling plans
D10	Repair money flow	Marketplace-capable PSP, explicit merchant/counterparty and fee bearer	PSP integration
D11	Voice retention	Editable transcript; temporary audio deleted ≤24h after transcription, unless approved exception	Personal-data processing
D12	Languages	IT/EN/BN tested first; ES prepared; no untested “all languages” guarantee	QA and pilot
D13	Qualification policy	Country/work-type matrix and evidence review, not category badge alone	Dispatch
D14	Matching commercial role	Organic launch; separate disclosed sponsorship only later	Ranking implementation
D15	Core plan limits	Publish fair use; no unlimited AI telephone included by assumption	Pricing/plan screen

ID	Decision	Proposed default / current posture	Needed before
D16	Custody scope	Enable pickup/return only when actual transport and liability process is staffed	Mode activation
D17	Cancellation/diagnosis/refunds	Written country-specific fees, consent and settlement rules	Customer booking
D18	Privacy and retention	Per-data-class schedule, vendor processing region/contracts and legal holds	Real user data
D19	Financial documents	Accountant-approved issuer, numbering, tax and e-invoice adapter per market	Invoice issuance
D20	Historical repository reuse	Audit source instead of treating old table counts/version labels as readiness proof	Estimates/schema implementation

Resolved priority conflicts

Latest pricing replaces old lead packages, free leads/referrals or per-ticket percentages unless separately reapproved. Retain the eight-category taxonomy rather than losing appliances. Retain PostgreSQL/PostGIS instead of reverting to MySQL. Preserve the artistic line/cream/ink direction from the reviewed PDF and handoff. New speech-to-text rule replaces older voice-note mockups. Shared-language preference is retained; nationality/culture matching and hidden paid ranking are replaced by the proposed suitability and transparency policy, awaiting owner review.

2. Market activation checklist

This is a product readiness checklist, not a legal opinion or a universal list of obligations. A named local reviewer must confirm the applicable requirements for the actual work, business model and jurisdiction. Do not paste a generic global policy into the app.

Area	Italy / European pilot	Selected US market
Trade scope	Work-type eligibility, required person/business qualifications and checks; F-gas verification where applicable	State/local trade licensing and work-scope verification; selected state/city first
Business/counterparty	Provider identity, entity/tax details, contracting relationship and insurance	Entity/provider onboarding, licensing, insurance and independent-contractor/marketplace review
Privacy	Applicable privacy notice, processing basis, rights, retention, vendors and cross-border processing review	Applicable state/federal privacy, communications and retention review
Consumer terms	Price disclosure, cancellation/withdrawal where applicable, urgent work consent, warranty/complaints	Applicable price/cancellation/repair warranty and consumer protection review
Money	PSP country support, onboarding, tax documents, refund/payout responsibilities	PSP/state coverage, fee allocation, tax/reporting and dispute model
Invoices	Approved issue/credit/numbering and any required local electronic workflows	Applicable sales tax/invoice/receipt treatment by work and jurisdiction
Emergency content	Locally approved scripts and official routes; no false emergency-service promise	Local official routes and appropriate scripts; no automatic reuse of Italian content

Area	Italy / European pilot	Selected US market
Calls/location	Approved disclosure, recording policy, active-job location notice	State-sensitive call/recording review and clear tracking notice

Sources W5–W8 are starting official resources, not evidence that Riparo is already compliant or eligible in a country. Complete specialist review before activating restricted categories or a new jurisdiction.

3. Acceptance scenarios

These are test specifications for developers/QA, not claims that tests were executed on an application in this documentation task.

Test	Given / action	Expected outcome
T01	Customer speaks Bangla about an appliance, edits transcription	Corrected original text persists; source revision and language retained
T02	Microphone permission refused	Typing/file selection remain usable; draft preserved
T03	Transcription/model timeout	Clear retry/edit/manual route; no fabricated summary or dispatch
T04	AI detects uncertain brand/model	Customer sees uncertainty and can select unknown; no forced invented identifier
T05	Customer confirms summary, then materially changes category	Confirmed revision invalidated; affected offers close/reconfirm
T06	Hazard phrase/rule triggered in intake or chat	Ordinary booking interrupted; approved local escalation; AI cannot override
T07	Provider lacks required valid scope	Excluded from offer/assignment regardless of proximity or payment plan
T08	Shop qualifies but assigned worker does not	Assignment prevented unless approved supervision policy actually allows it
T09	Ready part-time professional with no ratings	Eligible during declared hours; neutral prior and exploration opportunity
T10	Language preference unavailable	User informed; translated options remain; no ethnicity/nationality filtering
T11	Provider responds available	Customer notified; job not booked automatically; fee not charged
T12	Customer app background/muted	Visible/push state where permitted; no false guaranteed ringing promise
T13	Two booking commands/retries for one request	Exactly one active booking; idempotent result or clear conflict
T14	Two customers select same single-capacity worker/time	Only valid capacity commits; other customer can reselect
T15	Offer expires during booking	Transaction rejects stale response and releases hold safely

Test	Given / action	Expected outcome
T16	Queue restarts after booking commit	Outbox retries notify once effectively; booking remains durable
T17	Technician A opens shop B job/media/socket/export	Access denied without leaking private resource existence
T18	RAG query from shop A	No shop B CRM/manual-job documents retrieved
T19	Chat says €85, model translates numerals incorrectly	Structured quote remains €85; critical mismatch triggers correction, not auto approval
T20	Quote v1 approved; provider submits extra cost in v2	Extra work blocked pending v2 approval; v1 evidence retained
T21	Customer rejects diagnosis quote	Repair stops; only previously accepted diagnosis/return charges remain eligible
T22	Pickup repair completed but item not returned	Job shows ready/return pending; custody not falsely closed
T23	Wrong/damaged item at handover	Condition/accessory evidence and incident route; no silent acknowledgement
T24	Browser redirects to payment success without valid webhook	Payment stays pending until authoritative verification/reconciliation
T25	Same PSP webhook replayed/out of order	One financial effect; no state regression or duplicate refund
T26	First customer with shop A completes two jobs concurrently	One relationship and at most one acquisition debit
T27	Same customer returns to another A employee/branch	Zero further acquisition fee
T28	Same customer first completes work with independent B	One eligible fee for B; A history unchanged
T29	Walk-in contact/manual shop job completed	No marketplace acquisition debit
T30	Acquisition fee credited/refunded	Append-only credit; relationship persists; no accidental automatic re-charge
T31	Subscription unpaid during active customer repair	Grace/restriction policy applies; records and active support remain reachable
T32	Location stops updating or job ends	Stale marker/last update shown; sharing expires and further unauthorized posts rejected
T33	Schedule crosses DST change	Clear timezone/offset confirmation; intended slot preserved without collision
T34	Customer requests erasure with open legal/accounting retention	Process rights workflow; minimize retained data and explain approved exception
T35	New private upload with forged MIME/oversized media	Reject/quarantine safely; no public exposure or executable content

Test	Given / action	Expected outcome
T36	Provider certificate expires after offer but before booking	Recheck prevents invalid booking/assignment; support notified
T37	Screen-reader/keyboard/large-text customer approves quote	Accessible exact amount/version/action; focus and layout remain usable
T38	Account/organization merge reveals duplicate relationship	Audited canonicalization and adjustment; historical ledger not erased
T39	Database/Redis outage and restore drill	No invented state; recovery follows runbook; measured RPO/RTO recorded
T40	Inbound AI caller mentions emergency or asks for discount/payment	Approved handoff/safety behavior; no unauthorized booking, price promise or charge

For each implemented requirement, the team records covered test IDs, real command/results, environment and evidence link. Passing synthetic fixtures does not establish operational or legal readiness.

4. Project sources reviewed

The latest founder message dated 18 October 2026 is the primary authority for category intent, transcription behavior, provider/customer acceptance and the €9 plus once-per-customer fee model.

Current files read for this package: IMPLEMENTATION_STATUS.md (complete), README(1).md (complete), FINAL_DESIGN_AND_DEVELOPMENT_HANDOFF(1).md (design and core flow sections), Riparo_Italy_Pilot_Deep_Analysis_and_PRD.docx (opening analysis/requirements sections), and Repair App Wireframes(1).pdf (page 1 text and visual inspection). Prior conversation retrieval supplied additional project decisions. This is a targeted consolidation, not a claim to have read every account conversation, every PDF or the actual production source.

The supplied wireframe's arrival-code wording is ambiguous. This package standardizes customer-held code → technician validates on arrival. Older free-lead/referral offers and example price ranges are not automatically carried into the new commercial policy. Prior status/README files disagree on engine/framework major versions and describe incomplete integrations, so actual repository/runtime audit is mandatory.

5. Public references checked on 18 October 2026

These links support candidate selection and review checklists. Recheck version, tariff, licence and country availability at implementation; this document does not freeze a third party's future terms.

- W1 — SYSTRAN faster-whisper official repository: https://github.com/SYSTRAN/faster-whisper

- W2 — Qwen3-8B official model card: https://huggingface.co/Qwen/Qwen3-8B

- W3 — Qwen3-Embedding-0.6B official model card: https://huggingface.co/Qwen/Qwen3-Embedding-0.6B

- W4 — Gemini Developer API official pricing and data-use distinctions: https://ai.google.dev/gemini-api/docs/pricing

- W5 — Stripe Connect documentation, marketplace payment capabilities: https://docs.stripe.com/connect

- W6 — European Commission data protection overview: https://commission.europa.eu/law/law-topic/data-protection/data-protection-explained_en

- W7 — Italian F-gas portal: https://www.fgas.it/

- W8 — US Small Business Administration business launch guidance: https://www.sba.gov/counseling/launch-your-business/

6. Approval and next deliverables

Review D02–D10 first because they affect architecture, transactions and pricing. Once the requirements are accepted, produce the physical ERD/data dictionary, SQL/migrations and representative artistic UI screens. After those are reviewed, expand the full screen set and implement the first vertical slice. This ordering follows the founder's request to settle requirements before final database and mobile design.

Codex starter prompt

Copy the following into the coding session and attach all Markdown files and the actual repository.

> You are implementing Riparo / Repairmen. Read this package in the order in 00_READ_ME_FIRST.md, then read repository AGENTS.md and actual source. The package is a review baseline: latest founder decisions override older demos. First audit existing code, tests, schema and adapters, and produce a requirement-to-code reuse map. Distinguish verified code, demo-only code, missing integrations and untested claims. Do not implement unapproved policy choices as settled facts. Start from web/PWA first, mobile-first UX, Laravel modular backend, PostgreSQL/PostGIS, Redis and Next.js. Preserve the approved stack and existing working code. After requirements approval, produce a physical schema/migration plan before applying database changes. Enforce: tenant/participant isolation; one active booking per request; exact-version quote approval; one acquisition fee per canonical customer and provider organization; no repeat fee across shop staff/branches; deterministic eligibility/safety; idempotent webhooks/outbox; private media; voice input converted to editable text before send. AI cannot set binding prices, override permissions or charge payments. Work one reviewed feature increment at a time. Cite requirement IDs, define API/state/data effects, implement, run meaningful PostgreSQL/integration tests, demonstrate UI states and update traceability. Never claim a mock is production. Do not deploy production, run destructive migrations, send messages or introduce charges without the coding session's explicit scope/authority. Report concrete blockers and continue independent authorized work.

Suggested first tasks

1. Repository audit, version matrix, source reuse map and missing decisions.
2. Review logical model and generate proposed physical ERD/schema, without destructive execution.
3. After approval, tenant/auth foundation and migration tests.
4. One end-to-end ordinary repair vertical slice, including transcript confirmation, provider choice and quote approval.
5. Concurrency, billing, privacy, failure and real-provider integration gates before production.